

Lecture 02 - Rendering Pipeline

This lecture turns the vague idea of rendering into a concrete sequence. It is the most important debugging map in the course: if an image is wrong, the pipeline tells you where the error could have entered.

Source chunk	course_text_parts/03_lectures/02_rendering-pipeline.txt
Extracted slide pages	94
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Untitled slide
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Untitled slide. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Untitled slide' into a causal sentence instead of repeating the slide title?

Page 2	Object-based Rendering
Source cue	- Image generation done through rasterization / - Scene objects formed by 3D models (primitives) / are transformed through successive coordinate systems / - Primitives are rasterized into fragments, which are pixel predecessors / - A subset of the fragments becomes pixels
Commentary	This slide is about Object-based Rendering. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: - Image generation done through rasterization / - Scene objects formed by 3D models (primitives) / are transformed through successive coordinate systems / - Primitives are rasterized into fragments, which are pixel predecessors / - A subset of the fragments becomes pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.

Check yourself	Can you name the input and output representation for 'Object-based Rendering' in the rendering pipeline?
Page 3	2.1 Rendering Process
Source cue	2.2 OpenGL Overview / 2.3 OpenGL Rendering / 2.4 OpenGL Rendering Pipeline / 2.5 Fragment Tests and Operations / 2.6 Framebuffer
Commentary	This slide is about 2.1 Rendering Process. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: 2.2 OpenGL Overview / 2.3 OpenGL Rendering / 2.4 OpenGL Rendering Pipeline / 2.5 Fragment Tests and Operations / 2.6 Framebuffer
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for '2.1 Rendering Process' in the rendering pipeline?
Page 4	2.1 Rendering Process
Source cue	From 3D models over fragments to pixels
Commentary	This slide is about 2.1 Rendering Process. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: From 3D models over fragments to pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for '2.1 Rendering Process' in the rendering pipeline?
Page 5	Geometry-based Rendering 1/2
Source cue	CPU Application GPU Rendering Display Hardware / Rendering / 3D Model Framebuffer / Process / Video
Commentary	This slide is about Geometry-based Rendering 1/2. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: CPU Application GPU Rendering Display Hardware / Rendering / 3D Model Framebuffer / Process / Video
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'Geometry-based Rendering 1/2' in the rendering pipeline?
Page 6	Geometry-based Rendering 2/2
Source cue	- The rendering process is realized through the rendering pipeline / - Renders 3D models to 2D images / - Conceptually, the rendering pipeline consists of two GPU stages / - Geometry stage – processes geometric primitives / - Rasterization stage – processes fragments and pixels

Commentary	This slide is about Geometry-based Rendering 2/2. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: - The rendering process is realized through the rendering pipeline / - Renders 3D models to 2D images / - Conceptually, the rendering pipeline consists of two GPU stages / - Geometry stage – processes geometric primitives / - Rasterization stage – processes fragments and pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'Geometry-based Rendering 2/2' in the rendering pipeline?

Page 7	Application Stage
Source cue	- Specification of the model elements and the scene / as well as interaction capabilities / - Scene modeling: arrangement and attribution of the scene objects / - Calculation of animation paths / - Collision detection
Commentary	This slide is about Application Stage. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Specification of the model elements and the scene / as well as interaction capabilities / - Scene modeling: arrangement and attribution of the scene objects / - Calculation of animation paths / - Collision detection
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Application Stage' into a causal sentence instead of repeating the slide title?

Page 8	Geometry Stage
Source cue	$x \times x$ / - Geometric 3D transformation / Model Coordinate System World Coordinate System / - Alignment of individual scene objects in the overall scene / - Basic transformations: translation, scaling, rotation
Commentary	This slide is about Geometry Stage. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: $x \times x$ / - Geometric 3D transformation / Model Coordinate System World Coordinate System / - Alignment of individual scene objects in the overall scene / - Basic transformations: translation, scaling, rotation
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Geometry Stage'?

Page 9	Rasterization Stage
Source cue	- Conversion of primitives into fragments (rasterization) / - Discretization takes place in relation to a target grid / - Rasterization algorithms optimized for certain primitives / (e.g., lines, triangles) / - After rasterization per-fragment tests are performed

Commentary	This slide is about Rasterization Stage. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Conversion of primitives into fragments (rasterization) / - Discretization takes place in relation to a target grid / - Rasterization algorithms optimized for certain primitives / (e.g., lines, triangles) / - After rasterization per-fragment tests are performed
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Rasterization Stage'?
Page 10	2.2 OpenGL Overview
Source cue	Library for real-time computer graphics
Commentary	This slide is about 2.2 OpenGL Overview. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Library for real-time computer graphics
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2.2 OpenGL Overview'?
Page 11	OpenGL History
Source cue	- Unified graphics software interface was missing / as knowledge channeled into proprietary software packages / - HP's starbase / - SGI's Graphics Library (GL) / - Some standardization efforts existed
Commentary	This slide is about OpenGL History. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Unified graphics software interface was missing / as knowledge channeled into proprietary software packages / - HP's starbase / - SGI's Graphics Library (GL) / - Some standardization efforts existed
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL History'?
Page 12	OpenGL
Source cue	- OpenGL (Open Graphics Library) / - Platform independent 2D and 3D graphics API / - API specification is provided by the Khronos Group / - Technical implementation / - Rendering pipeline consisting of static and programmable pipeline stages

Commentary	This slide is about OpenGL. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: - OpenGL (Open Graphics Library) / - Platform independent 2D and 3D graphics API / - API specification is provided by the Khronos Group / - Technical implementation / - Rendering pipeline consisting of static and programmable pipeline stages
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'OpenGL' in the rendering pipeline?
Page 13	OpenGL vs. Vulkan
Source cue	- How about Vulkan? / - More modern than OpenGL / - Designed for high performance and low-level GPU control / - Better suited for complex, large-scale rendering engines / - Vulkan also harder to learn and requires more setup/code
Commentary	This slide is about OpenGL vs. Vulkan. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - How about Vulkan? / - More modern than OpenGL / - Designed for high performance and low-level GPU control / - Better suited for complex, large-scale rendering engines / - Vulkan also harder to learn and requires more setup/code
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL vs. Vulkan'?
Page 14	OpenGL Libraries
Source cue	- Different tastes of OpenGL / - OpenGL / - OpenGL ES - limited version for embedded systems / - WebGL - Javascript connection for Web Development / (functionality similar to OpenGL ES)
Commentary	This slide is about OpenGL Libraries. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Different tastes of OpenGL / - OpenGL / - OpenGL ES - limited version for embedded systems / - WebGL - Javascript connection for Web Development / (functionality similar to OpenGL ES)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Libraries'?
Page 15	OpenGL Functionality
Source cue	- Functional API / - Objects, scenes and images are described by a sequence of commands / (functional approach, low-level) / - Contrast: scene or object description (declarative approach, high-level) / - OpenGL context

Commentary	This slide is about OpenGL Functionality. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Functional API / - Objects, scenes and images are described by a sequence of commands / (functional approach, low-level) / - Contrast: scene or object description (declarative approach, high-level) / - OpenGL context
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Functionality'?

Page 16	OpenGL Context
Source cue	- Context is created and modified by the application, / and contains all the state variables of the OpenGL engine / - Any number of contexts per application possible, / but at most one active context per application / - Context is generally tied to an output window
Commentary	This slide is about OpenGL Context. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Context is created and modified by the application, / and contains all the state variables of the OpenGL engine / - Any number of contexts per application possible, / but at most one active context per application / - Context is generally tied to an output window
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Context'?

Page 17	OpenGL Rendering Process
Source cue	- Initiation from the developer's perspective / - Creation of the output window / - Creation of the context / - Specification of the scene geometry / - Uploading of the scene geometry to the GPU
Commentary	This slide is about OpenGL Rendering Process. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: - Initiation from the developer's perspective / - Creation of the output window / - Creation of the context / - Specification of the scene geometry / - Uploading of the scene geometry to the GPU
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'OpenGL Rendering Process' in the rendering pipeline?

Page 18	Window Creation with GLFW 1/2
Source cue	<code>#include <GLFW/glfw3.h> / #include <stdlib.h> / #include <stdio.h> / static void error_callback(int error, const char* description) { / fputs(description, stderr);</code>

Commentary	This slide is about Window Creation with GLFW 1/2. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: <code>#include <GLFW/glfw3.h> / #include <stdlib.h> / #include <stdio.h> / static void error_callback(int error, const char* description) { / fputs(description, stderr);</code>
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Window Creation with GLFW 1/2' into a causal sentence instead of repeating the slide title?
Page 19	Window Creation with GLFW 2/2
Source cue	<code>int main(void){ / GLFWwindow* window; / glfwSetErrorCallback(error_callback); / if (!glfwInit()) exit(EXIT_FAILURE); / window = glfwCreateWindow(640, 480, "Simple example", NULL, NULL);</code>
Commentary	This slide is about Window Creation with GLFW 2/2. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: <code>int main(void){ / GLFWwindow* window; / glfwSetErrorCallback(error_callback); / if (!glfwInit()) exit(EXIT_FAILURE); / window = glfwCreateWindow(640, 480, "Simple example", NULL, NULL);</code>
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Window Creation with GLFW 2/2' into a causal sentence instead of repeating the slide title?
Page 20	OpenGL Naming Conventions
Source cue	- Function-Prefix "gl" / - glClear, ... / - Constant-Prefix "GL" / - GL_POLYGON, GL_LINES, ... / - Parameter types are explicitly encoded in function names
Commentary	This slide is about OpenGL Naming Conventions. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Function-Prefix "gl" / - glClear, ... / - Constant-Prefix "GL" / - GL_POLYGON, GL_LINES, ... / - Parameter types are explicitly encoded in function names
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Naming Conventions'?
Page 21	2.3 OpenGL Rendering
Source cue	Transfer of 3D models to the GPU
Commentary	This slide is about 2.3 OpenGL Rendering. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Transfer of 3D models to the GPU

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2.3 OpenGL Rendering'?
Page 22	OpenGL Rendering (until V2.1)
Source cue	- Rendering is done through Immediate Mode / void renderScene() { / glColor3f(0.0f, 1.0f, 0.0f); // drawing color becomes green / glBegin(GL_POLYGON); // begin primitive / glVertex2f(-0.5f,-0.5f); // 1. vertex
Commentary	This slide is about OpenGL Rendering (until V2.1). Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Rendering is done through Immediate Mode / void renderScene() { / glColor3f(0.0f, 1.0f, 0.0f); // drawing color becomes green / glBegin(GL_POLYGON); // begin primitive / glVertex2f(-0.5f,-0.5f); // 1. vertex
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Rendering (until V2.1)'?
Page 23	OpenGL Rendering (since V3.0)
Source cue	- Using Vertex Array Objects (VAOs) / - Vertex specification in CPU memory / - Vertex data transfer from CPU to GPU / - Memory layout description of the vertex data / - Number of vertices, number of components (x, y, z, ...)
Commentary	This slide is about OpenGL Rendering (since V3.0). Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Using Vertex Array Objects (VAOs) / - Vertex specification in CPU memory / - Vertex data transfer from CPU to GPU / - Memory layout description of the vertex data / - Number of vertices, number of components (x, y, z, ...)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Rendering (since V3.0)'?
Page 24	1. Vertex Specification
Source cue	- Vertex data is specified as a C array // vertices of two triangles / GLfloat vertices[6][2] = { // 6 vertices with two components each (x,y) / {0.0, 1.0}, {1.0, 1.0}, {1.0, 0.0}, // 1. triangle / {0.5, 0.5}, {1.0, 0.3}, {0.7, 1.0} // 2. triangle

Commentary	This slide is about 1. Vertex Specification. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Vertex data is specified as a C array / // vertices of two triangles / GLfloat vertices[6][2] = { // 6 vertices with two components each (x,y) / {0.0, 1.0}, {1.0, 1.0}, {1.0, 0.0}, // 1. triangle / {0.5, 0.5}, {1.0, 0.3}, {0.7, 1.0} // 2. triangle
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '1. Vertex Specification'?

Page 25	2. Vertex Data Transfer
Source cue	- VBO allocation and data transfer via / - glBufferData(GL_ARRAY_BUFFER, size, data, GL_STATIC_DRAW); / - data – address in main memory (i.e., vertices or &vertices[0]) / - size – size of data in number of bytes / (i.e. sizeof(vertices) or numVertices*sizeof(GLfloat)*2)
Commentary	This slide is about 2. Vertex Data Transfer. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - VBO allocation and data transfer via / - glBufferData(GL_ARRAY_BUFFER, size, data, GL_STATIC_DRAW); / - data – address in main memory (i.e., vertices or &vertices[0]) / - size – size of data in number of bytes / (i.e. sizeof(vertices) or numVertices*sizeof(GLfloat)*2)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2. Vertex Data Transfer'?

Page 26	3. Memory Layout Description
Source cue	- Memory layout is described by the following function / - glVertexAttribPointer(attribIndex, size, type, normalized, stride, offset); / - attribIndex – attribute number (0 for position, later more ...) / - size – number of components per vertex (2 for xy, 3 for xyz) / - type – base type (typically GL_FLOAT or GL_DOUBLE)
Commentary	This slide is about 3. Memory Layout Description. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Memory layout is described by the following function / - glVertexAttribPointer(attribIndex, size, type, normalized, stride, offset); / - attribIndex – attribute number (0 for position, later more ...) / - size – number of components per vertex (2 for xy, 3 for xyz) / - type – base type (typically GL_FLOAT or GL_DOUBLE)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '3. Memory Layout Description'?

Page 27	4. VAO Drawing 1/2
Source cue	- Drawing is done with the command / - <code>glDrawArrays(mode, first, count);</code> / - mode – specifies the primitive type (e.g., <code>GL_TRIANGLES</code> or <code>GL_LINES</code>) / - From all vertices, the next count vertices are drawn from index first / - Actual vertex data no longer needs to be specified
Commentary	This slide is about 4. VAO Drawing 1/2. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Drawing is done with the command / - <code>glDrawArrays(mode, first, count);</code> / - mode – specifies the primitive type (e.g., <code>GL_TRIANGLES</code> or <code>GL_LINES</code>) / - From all vertices, the next count vertices are drawn from index first / - Actual vertex data no longer needs to be specified
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '4. VAO Drawing 1/2'?

Page 28	4. VAO Drawing 2/2
Source cue	- There is a small number of primitive types / - Points / - Lines, Line Strips, Line Loops / - Triangles, Triangle Strips, Triangle Fans, Patches / v2 v2 v2
Commentary	This slide is about 4. VAO Drawing 2/2. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - There is a small number of primitive types / - Points / - Lines, Line Strips, Line Loops / - Triangles, Triangle Strips, Triangle Fans, Patches / v2 v2 v2
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '4. VAO Drawing 2/2'?

Page 29	5. GPU Memory Release
Source cue	- After usage VAO and VBO should be deleted from GPU memory / - <code>glDeleteBuffers(...);</code> / - <code>glDeleteVertexArrays(...);</code>
Commentary	This slide is about 5. GPU Memory Release. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - After usage VAO and VBO should be deleted from GPU memory / - <code>glDeleteBuffers(...);</code> / - <code>glDeleteVertexArrays(...);</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '5. GPU Memory Release'?

Page 30	Rendering Initialization 1/2
Source cue	<code>static void init_render_scene(void) { / static const GLfloat vertices[6][2] = { / {0.0f, 1.0f}, {1.0f, 1.0f}, {1.0f, 0.0f}, /* 1st triangle */ / {0.5f, 0.5f}, {1.0f, 0.3f}, {0.7f, 1.0f} /* 2nd triangle */ / static const char* vs_src =</code>

Commentary	This slide is about Rendering Initialization 1/2. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: <pre>static void init_render_scene(void) { / static const GLfloat vertices[6][2] = { / {0.0f, 1.0f}, {1.0f, 1.0f}, {1.0f, 0.0f}, /* 1st triangle */ / {0.5f, 0.5f}, {1.0f, 0.3f}, {0.7f, 1.0f} /* 2nd triangle */ / static const char* vs_src =</pre>
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Rendering Initialization 1/2'?
Page 31	Rendering Initialization 2/2
Source cue	<pre>program = create_program(vs_src, fs_src); / glGenVertexArrays(1, &vao); / glGenBuffers(1, &vbo); / glBindVertexArray(vao); / glBindBuffer(GL_ARRAY_BUFFER, vbo);</pre>
Commentary	This slide is about Rendering Initialization 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: <pre>program = create_program(vs_src, fs_src); / glGenVertexArrays(1, &vao); / glGenBuffers(1, &vbo); / glBindVertexArray(vao); / glBindBuffer(GL_ARRAY_BUFFER, vbo);</pre>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Rendering Initialization 2/2'?
Page 32	Rendering
Source cue	<pre>static void render_scene(void) { / glClear(GL_COLOR_BUFFER_BIT); / glUseProgram(program); / glBindVertexArray(vao); / glDrawArrays(GL_TRIANGLES, 0, 6);</pre>
Commentary	This slide is about Rendering. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: <pre>static void render_scene(void) { / glClear(GL_COLOR_BUFFER_BIT); / glUseProgram(program); / glBindVertexArray(vao); / glDrawArrays(GL_TRIANGLES, 0, 6);</pre>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Rendering'?
Page 33	Additional Attributes
Source cue	- Additional attributes are necessary for drawing / - Color, normals, texture coordinates, ... / - There are two ways to assign these attributes to the vertex data / - Attributes are bound to the same VAO in separate VBO / - Attributes are stored interleaved with vertex data,

Commentary	This slide is about Additional Attributes. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Additional attributes are necessary for drawing / - Color, normals, texture coordinates, ... / - There are two ways to assign these attributes to the vertex data / - Attributes are bound to the same VAO in separate VBO / - Attributes are stored interleaved with vertex data,
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Additional Attributes'?

Page 34	2.4 OpenGL Rendering Pipeline
Source cue	Programmable and static stages
Commentary	This slide is about 2.4 OpenGL Rendering Pipeline. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: Programmable and static stages
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for '2.4 OpenGL Rendering Pipeline' in the rendering pipeline?

Page 35	The Rendering Pipeline 1/2
Source cue	Scene Description Raster Image / Geometry Fragment / Rasterization / Processing Processing / Vertices Primitives Fragments Pixels
Commentary	This slide is about The Rendering Pipeline 1/2. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: Scene Description Raster Image / Geometry Fragment / Rasterization / Processing Processing / Vertices Primitives Fragments Pixels
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'The Rendering Pipeline 1/2' in the rendering pipeline?

Page 36	The Rendering Pipeline 2/2
Source cue	- Sequence of fixed and programmable stages Programmable / Vertex Processing / - Process is influenced by context and scene objects / Context Fixed Function / - Programmable levels are programmed
Commentary	This slide is about The Rendering Pipeline 2/2. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: - Sequence of fixed and programmable stages Programmable / Vertex Processing / - Process is influenced by context and scene objects / Context Fixed Function / - Programmable levels are programmed
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.

Check yourself	Can you name the input and output representation for 'The Rendering Pipeline 2/2' in the rendering pipeline?
Page 37	Shader Programming – History 1/2
Source cue	- There is special graphics hardware (dedicated) / and general graphics hardware (general purpose) / - Special graphics hardware / - Volume Rendering: VolumePro / - Game consoles: Custom NVIDIA chips in Xbox
Commentary	This slide is about Shader Programming – History 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - There is special graphics hardware (dedicated) / and general graphics hardware (general purpose) / - Special graphics hardware / - Volume Rendering: VolumePro / - Game consoles: Custom NVIDIA chips in Xbox
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Programming – History 1/2'?
Page 38	Shader Programming – History 2/2
Source cue	- Various vendor-dependent extensions are introduced / to provide extensibility for general graphics hardware / - Texture shaders and register combiners allowed modification / through the application of textures / - Applicability is very limited by the possibilities of graphics hardware
Commentary	This slide is about Shader Programming – History 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Various vendor-dependent extensions are introduced / to provide extensibility for general graphics hardware / - Texture shaders and register combiners allowed modification / through the application of textures / - Applicability is very limited by the possibilities of graphics hardware
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Programming – History 2/2'?
Page 39	Assembler Shader
Source cue	- First OpenGL extensions support shader programming in assembler code / <code>!! ARBvp1.0 / ATTRIB iPos = vertex.position ; / ATTRIB iNormal = vertex.normal ; / ...</code>
Commentary	This slide is about Assembler Shader. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - First OpenGL extensions support shader programming in assembler code / <code>!! ARBvp1.0 / ATTRIB iPos = vertex.position ; / ATTRIB iNormal = vertex.normal ; / ...</code>

Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Assembler Shader'?
Page 40	High-Level Shading Languages 1/2
Source cue	- Assembler code is hard to read and hard to maintain / - High-level languages for shading are specified in C-like syntax / - Source code is translated by the compiler into appropriate ASM statements / - Using shaders after initial binding automatically when rendering geometry / - Interface to OpenGL by passing appropriate variables or changing the OpenGL
Commentary	This slide is about High-Level Shading Languages 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Assembler code is hard to read and hard to maintain / - High-level languages for shading are specified in C-like syntax / - Source code is translated by the compiler into appropriate ASM statements / - Using shaders after initial binding automatically when rendering geometry / - Interface to OpenGL by passing appropriate variables or changing the OpenGL
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'High-Level Shading Languages 1/2'?
Page 41	High-Level Shading Languages 2/2
Source cue	- Existing high level languages for shader development / - CG - C for Graphics by NVIDIA / - CG was introduced in 2003 by NVIDIA as a platform independent solution / (in practice rather semi-platform independent) / - HLSL - High Level Shading Language from Microsoft
Commentary	This slide is about High-Level Shading Languages 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Existing high level languages for shader development / - CG - C for Graphics by NVIDIA / - CG was introduced in 2003 by NVIDIA as a platform independent solution / (in practice rather semi-platform independent) / - HLSL - High Level Shading Language from Microsoft
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'High-Level Shading Languages 2/2'?
Page 42	Shader Types

Source cue	- Pipeline concept has been kept in place / and shaders are responsible for different stages / - First, geometry level programming was done by vertex shaders / - Vertex shaders allow modifiability of vertex transformations and lighting / - After that, there was an extension for image-based algorithms by fragment
Commentary	This slide is about Shader Types. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Pipeline concept has been kept in place / and shaders are responsible for different stages / - First, geometry level programming was done by vertex shaders / - Vertex shaders allow modifiability of vertex transformations and lighting / - After that, there was an extension for image-based algorithms by fragment
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Types'?

Page 43	Vertex Shader Concepts
Source cue	- Vertex shaders enable the programmable modification of processed geometry / - Possibilities of a vertex shader / Programmable / - Geometric transformation of vertices and normals / Vertex Processing
Commentary	This slide is about Vertex Shader Concepts. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Vertex shaders enable the programmable modification of processed geometry / - Possibilities of a vertex shader / Programmable / - Geometric transformation of vertices and normals / Vertex Processing
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Concepts'?

Page 44	Vertex Shader Examples 1/2
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Vertex Shader Examples 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Examples 1/2'?

Page 45	Vertex Shader Examples 2/2
---------	----------------------------

Source cue	- Morphing / - Shadow volume calculation for shadow effects / [ID Software's Doom III]
Commentary	This slide is about Vertex Shader Examples 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Morphing / - Shadow volume calculation for shadow effects / [ID Software's Doom III]
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Examples 2/2'?

Page 46	Vertex Shader Coordinate Systems 1/2
Source cue	- Vertex shaders are executed for each vertex / - Vertex shaders calculate coordinates in clipping coordinates as output / - Vertex shaders can not / - Access adjacent vertices / - Change the number of vertices (i.e., generate or discard vertices)
Commentary	This slide is about Vertex Shader Coordinate Systems 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Vertex shaders are executed for each vertex / - Vertex shaders calculate coordinates in clipping coordinates as output / - Vertex shaders can not / - Access adjacent vertices / - Change the number of vertices (i.e., generate or discard vertices)
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Coordinate Systems 1/2'?

Page 47	Vertex Shader Coordinate Systems 2/2
Source cue	Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform
Commentary	This slide is about Vertex Shader Coordinate Systems 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Coordinate Systems 2/2'?

Page 48	Local Space
Source cue	- Coordinates of your object relative to its local origin / - 3D vertex positions

Commentary	This slide is about Local Space. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Coordinates of your object relative to its local origin / - 3D vertex positions
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Local Space'?
Page 49	World Space
Source cue	- Coordinates in respect of a larger world / - Relative to some global origin of the world / - Together with many other objects
Commentary	This slide is about World Space. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Coordinates in respect of a larger world / - Relative to some global origin of the world / - Together with many other objects
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'World Space'?
Page 50	View Space
Source cue	- Coordinates as seen from the camera
Commentary	This slide is about View Space. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Coordinates as seen from the camera
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'View Space'?
Page 51	Clip Space
Source cue	- Clip coordinates range from -1 to 1 / - Projection (Orthographic, Perspective)
Commentary	This slide is about Clip Space. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Clip coordinates range from -1 to 1 / - Projection (Orthographic, Perspective)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Clip Space'?
Page 52	Screen Space
Source cue	- Transform to image coordinates with glViewport / - Results are sent to the rasterizer

Commentary	This slide is about Screen Space. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Transform to image coordinates with glViewport / - Results are sent to the rasterizer
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Screen Space'?
Page 53	Coordinate Systems
Source cue	Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform
Commentary	This slide is about Coordinate Systems. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Local Space World Space View Space Clip Space Screen Space / Viewport / ■ ■ ■ / ■■■■■■ ■■■■ ■■■■■■■■ Transform
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Coordinate Systems'?
Page 54	Vertex Shader Variables
Source cue	- Input types / - Uniforms: read only in VS (& FS) // Intrinsic vertex attributes / out gl_PerVertex { / - For example: light position or light color / vec4 gl_Position;
Commentary	This slide is about Vertex Shader Variables. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Input types / - Uniforms: read only in VS (& FS) // Intrinsic vertex attributes / out gl_PerVertex { / - For example: light position or light color / vec4 gl_Position;
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Vertex Shader Variables'?
Page 55	Fixed Function Vertex Postprocessing
Source cue	- Clipping / - Discard invisible geometries / Programmable / - Perspective Division / Vertex Processing
Commentary	This slide is about Fixed Function Vertex Postprocessing. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Clipping / - Discard invisible geometries / Programmable / - Perspective Division / Vertex Processing
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.

Check yourself	Can you explain how 'Fixed Function Vertex Postprocessing' changes positions before rasterization?
Page 56	Fixed Function Rasterization
Source cue	- Geometry is converted to raster coordinates / Programmable / Vertex Processing / Fixed Function / Primitives Fragments Vertex Postprocessing
Commentary	This slide is about Fixed Function Rasterization. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Geometry is converted to raster coordinates / Programmable / Vertex Processing / Fixed Function / Primitives Fragments Vertex Postprocessing
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Fixed Function Rasterization'?
Page 57	Fragment Shader Concept 1/2
Source cue	- Fragment shaders set the color of a pixel / - Possibilities of a fragment shader / Programmable / - Glare of textures / Vertex Processing
Commentary	This slide is about Fragment Shader Concept 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Fragment shaders set the color of a pixel / - Possibilities of a fragment shader / Programmable / - Glare of textures / Vertex Processing
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Fragment Shader Concept 1/2'?
Page 58	Fragment Shader Concept 2/2
Source cue	- Fragment shaders are executed for fragments (= pixel predecessors) / - Fragment shaders calculate the color and depth value of the current fragment / - No information about adjacent fragments available / - The number of fragments can be changed / - Fragment shaders can not generate fragments
Commentary	This slide is about Fragment Shader Concept 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Fragment shaders are executed for fragments (= pixel predecessors) / - Fragment shaders calculate the color and depth value of the current fragment / - No information about adjacent fragments available / - The number of fragments can be changed / - Fragment shaders can not generate fragments
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.

Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Fragment Shader Concept 2/2'?
Page 59	Shader Example
Source cue	layout (std140) uniform Matrices { Vertex Shader / mat4 projModelViewMatrix; / mat3 normalMatrix; /** Fragment Shader / }; * Set fragment color to red. / in vec3 position;
Commentary	This slide is about Shader Example. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: layout (std140) uniform Matrices { Vertex Shader / mat4 projModelViewMatrix; / mat3 normalMatrix; /** Fragment Shader / }; * Set fragment color to red. / in vec3 position;
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Example'?
Page 60	GLSL Shader Functions
Source cue	- Trigonometry functions / - radians, degrees, sin, cos, tan, asin, acos, atan / - Power calculations / - pow, exp2, log2, sqrt, inversesqrt / - General
Commentary	This slide is about GLSL Shader Functions. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Trigonometry functions / - radians, degrees, sin, cos, tan, asin, acos, atan / - Power calculations / - pow, exp2, log2, sqrt, inversesqrt / - General
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'GLSL Shader Functions'?
Page 61	Shader Usage
Source cue	- One or multiple shaders are attached to a shader program / Vertex Shader / glCreateShader glCreateProgram / Fragment Shader / glShaderSource glAttachShader glCreateShader
Commentary	This slide is about Shader Usage. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - One or multiple shaders are attached to a shader program / Vertex Shader / glCreateShader glCreateProgram / Fragment Shader / glShaderSource glAttachShader glCreateShader
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Usage'?
Page 62	Binding Shaders

Source cue	<pre>v = glCreateShader(GL_VERTEX_SHADER); // vertex shader / char* vs = readTextFile("vertex-shader.vert"); / const char* vv = vs; / glShaderSource(v, 1, &vv, NULL); / free(vs);</pre>
Commentary	This slide is about Binding Shaders. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: <pre>v = glCreateShader(GL_VERTEX_SHADER); // vertex shader / char* vs = readTextFile("vertex-shader.vert"); / const char* vv = vs; / glShaderSource(v, 1, &vv, NULL); / free(vs);</pre>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Binding Shaders'?

Page 63	Setting Uniforms
Source cue	- Uniforms are used to communicate between CPU and GPU / - Data is bound to uniform locations / - Must be updated after each linking process / <pre>GLint loc0 = glGetUniformLocation(p, "isoValue_"); / glUniform1f(loc0, 0.45f);</pre>
Commentary	This slide is about Setting Uniforms. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Uniforms are used to communicate between CPU and GPU / - Data is bound to uniform locations / - Must be updated after each linking process / <pre>GLint loc0 = glGetUniformLocation(p, "isoValue_"); / glUniform1f(loc0, 0.45f);</pre>
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Setting Uniforms' into a causal sentence instead of repeating the slide title?

Page 64	Shader Performance
Source cue	- Guidelines for efficient shaders / - Since fragment shaders are executed for a large number of fragments, / they should contain as few instructions as possible / - There should be as little as possible a change of the active shaders, / because they have their own state and, accordingly, the pipeline may need
Commentary	This slide is about Shader Performance. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Guidelines for efficient shaders / - Since fragment shaders are executed for a large number of fragments, / they should contain as few instructions as possible / - There should be as little as possible a change of the active shaders, / because they have their own state and, accordingly, the pipeline may need
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Performance'?

Page 65	Shader Development Environments 1/2
Source cue	- FX Composer, NVPerfHUD (NVIDIA) / - HLSL Shader IDE with performance analytics / www.developer.nvidia.com/page/tools.html / - RenderMonkey (ATI) / - HLSL, GLSL Shader IDE with performance analytics
Commentary	This slide is about Shader Development Environments 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - FX Composer, NVPerfHUD (NVIDIA) / - HLSL Shader IDE with performance analytics / www.developer.nvidia.com/page/tools.html / - RenderMonkey (ATI) / - HLSL, GLSL Shader IDE with performance analytics
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Development Environments 1/2'?
Page 66	Shader Development Environments 2/2
Source cue	- Shdr online GLSL editor: https://shdr.bkcore.com/
Commentary	This slide is about Shader Development Environments 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Shdr online GLSL editor: https://shdr.bkcore.com/
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Shader Development Environments 2/2'?
Page 67	2.5 Fragment Tests and Operations
Source cue	Towards the finishing line to the pixel
Commentary	This slide is about 2.5 Fragment Tests and Operations. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: Towards the finishing line to the pixel
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '2.5 Fragment Tests and Operations'?
Page 68	Fragment Tests and Operations 1/2
Source cue	- After fragments have been generated, they must go through the per-fragment / tests and operations / Programmable / - Only if all these tests are successful, / Vertex Processing

Commentary	This slide is about Fragment Tests and Operations 1/2. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - After fragments have been generated, they must go through the per-fragment / tests and operations / Programmable / - Only if all these tests are successful, / Vertex Processing
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Fragment Tests and Operations 1/2' into a causal sentence instead of repeating the slide title?

Page 69	Fragment Tests and Operations 2/2
Source cue	Fragment Pixel Ownership Scissor Alpha / Data Test Test Test / Depth Stencil / Framebuffer Dithering Blending / Test Test
Commentary	This slide is about Fragment Tests and Operations 2/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Fragment Pixel Ownership Scissor Alpha / Data Test Test Test / Depth Stencil / Framebuffer Dithering Blending / Test Test
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Fragment Tests and Operations 2/2'?

Page 70	Alpha Test
Source cue	- If framebuffer is an RGBA buffer, the alpha test can be performed / - The test must first be activated / glEnable(GL_ALPHA_TEST); / - The comparison function indicates which fragments survive the test / glAlphaFunc(func, value);
Commentary	This slide is about Alpha Test. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - If framebuffer is an RGBA buffer, the alpha test can be performed / - The test must first be activated / glEnable(GL_ALPHA_TEST); / - The comparison function indicates which fragments survive the test / glAlphaFunc(func, value);
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Alpha Test'?

Page 71	Stencil Test
Source cue	- Stencil Buffer / - Is a frame buffer component / - Is (indirectly) filled by drawing operations / - No special stencil drawing operations / - Stencil operation

Commentary	This slide is about Stencil Test. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Stencil Buffer / - Is a frame buffer component / - Is (indirectly) filled by drawing operations / - No special stencil drawing operations / - Stencil operation
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Stencil Test'?
Page 72	Depth Test
Source cue	- If the framebuffer has a depth buffer, the depth test can be performed / - The test must first be activated / <code>glEnable(GL_DEPTH_TEST)</code> ; / - The comparison is made with the depth value already in the buffer / - The comparison function indicates which fragments survive the test
Commentary	This slide is about Depth Test. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - If the framebuffer has a depth buffer, the depth test can be performed / - The test must first be activated / <code>glEnable(GL_DEPTH_TEST)</code> ; / - The comparison is made with the depth value already in the buffer / - The comparison function indicates which fragments survive the test
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Depth Test'?
Page 73	Image Composition and Mixture
Source cue	- Alpha values of the pixels / - control the combination of pixel values / - are specified as color components of color pixels / - $\mathbf{c} = [\mathbf{r}, \mathbf{g}, \mathbf{b}, \mathbf{a}]$, $\mathbf{c} \in [0,1]$ / - Applications are manifold
Commentary	This slide is about Image Composition and Mixture. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Alpha values of the pixels / - control the combination of pixel values / - are specified as color components of color pixels / - $\mathbf{c} = [\mathbf{r}, \mathbf{g}, \mathbf{b}, \mathbf{a}]$, $\mathbf{c} \in [0,1]$ / - Applications are manifold
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Image Composition and Mixture' into a causal sentence instead of repeating the slide title?
Page 74	Blending Application
Source cue	- Example application for transparency: drawing with semi-transparent brush

Commentary	This slide is about Blending Application. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Example application for transparency: drawing with semi-transparent brush
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Blending Application' into a causal sentence instead of repeating the slide title?

Page 75	Blending Process 1/2
Source cue	- Blending is an operation called per fragment / - Fragments are modified after rasterization, / but before writing to the framebuffer with a blending function / - Combination of a source fragment to be written with the destination pixel / in the frame buffer by using a blending function
Commentary	This slide is about Blending Process 1/2. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Blending is an operation called per fragment / - Fragments are modified after rasterization, / but before writing to the framebuffer with a blending function / - Combination of a source fragment to be written with the destination pixel / in the frame buffer by using a blending function
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Blending Process 1/2'?

Page 76	Blending Process 2/2
Source cue	- Source factor: ■ / - Destination factor: ■ / - Source fragment: [■ , ■ , ■ , ■] / ■,■ ■,■ ■,■ ■,■ ■ / - Destination pixel: [■ , ■ , ■ , ■]
Commentary	This slide is about Blending Process 2/2. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Source factor: ■ / - Destination factor: ■ / - Source fragment: [■ , ■ , ■ , ■] / ■,■ ■,■ ■,■ ■,■ ■ / - Destination pixel: [■ , ■ , ■ , ■]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Blending Process 2/2' into a causal sentence instead of repeating the slide title?

Page 77	Applying Blending
Source cue	- Switch on (or switch off) blending / - glEnable(GL_BLEND); / - glDisable(GL_BLEND); / - Defining Blending Factors / - glBlendFunc(srcFac, destFac);

Commentary	This slide is about Applying Blending. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Switch on (or switch off) blending / - glEnable(GL_BLEND); / - glDisable(GL_BLEND); / - Defining Blending Factors / - glBlendFunc(srcFac, destFac);
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Applying Blending' into a causal sentence instead of repeating the slide title?
Page 78	Blending Examples
Source cue	- Example 1: Copy source pixels and overwrite destination pixels / - glBlendFunc(GL_ONE, GL_ZERO); / - ■ = [■, ■, ■, ■] / ■■■■■■ ■, ■■■■■■ ■, ■■■■■■ ■, ■■■■■■ ■ / - Example 2: Alpha-based mixing of source pixels and destination pixels
Commentary	This slide is about Blending Examples. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Example 1: Copy source pixels and overwrite destination pixels / - glBlendFunc(GL_ONE, GL_ZERO); / - ■ = [■, ■, ■, ■] / ■■■■■■ ■, ■■■■■■ ■, ■■■■■■ ■, ■■■■■■ ■ / - Example 2: Alpha-based mixing of source pixels and destination pixels
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Blending Examples' into a causal sentence instead of repeating the slide title?
Page 79	2.6 Framebuffer
Source cue	Contains everything that is (in)visible on the screen
Commentary	This slide is about 2.6 Framebuffer. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Contains everything that is (in)visible on the screen
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '2.6 Framebuffer'?
Page 80	Viewport
Source cue	- Rectangular area in pixel grid, set explicitly / - Rendering operations are limited to this area / - All primitives that are completely or partially outside / are clipped at the viewport area (clipping)

Commentary	This slide is about Viewport. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Rectangular area in pixel grid, set explicitly / - Rendering operations are limited to this area / - All primitives that are completely or partially outside / are clipped at the viewport area (clipping)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Viewport' changes positions before rasterization?

Page 81	Framebuffer
Source cue	- Context-bound raster with a resolution and structure / specified by the application / - Physical realization on the GPU / - Logical subdivision / - Color memory
Commentary	This slide is about Framebuffer. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Context-bound raster with a resolution and structure / specified by the application / - Physical realization on the GPU / - Logical subdivision / - Color memory
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Framebuffer'?

Page 82	OpenGL Framebuffers 1/3
Source cue	- Typical OpenGL framebuffer configuration / 8bit Stencil Buffer / 32bit Depth Buffer / = 32bit Color Buffer
Commentary	This slide is about OpenGL Framebuffers 1/3. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Typical OpenGL framebuffer configuration / 8bit Stencil Buffer / 32bit Depth Buffer / = 32bit Color Buffer
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Framebuffers 1/3'?

Page 83	OpenGL Framebuffers 2/3
Source cue	- All buffers of a frame buffer have the same grid size / - Number of bitplanes per buffer can be configured individually / - Individual buffers are directly supported by the hardware / - Buffer identifier / - GL_COLOR_BUFFER_BIT

Commentary	This slide is about OpenGL Framebuffers 2/3. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - All buffers of a frame buffer have the same grid size / - Number of bitplanes per buffer can be configured individually / - Individual buffers are directly supported by the hardware / - Buffer identifier / - GL_COLOR_BUFFER_BIT
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Framebuffers 2/3'?
Page 84	OpenGL Framebuffers 3/3
Source cue	- Delete a buffer / <code>glClear(GLbitfield mask);</code> / - Set the initialization color for the color buffer / <code>glClearColor(1.0,1.0,1.0,1.0);</code> / <code>glClear(GL_COLOR_BUFFER_BIT);</code>
Commentary	This slide is about OpenGL Framebuffers 3/3. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Delete a buffer / <code>glClear(GLbitfield mask);</code> / - Set the initialization color for the color buffer / <code>glClearColor(1.0,1.0,1.0,1.0);</code> / <code>glClear(GL_COLOR_BUFFER_BIT);</code>
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'OpenGL Framebuffers 3/3'?
Page 85	Double Buffering
Source cue	- Use of two color buffers to avoid flicker effects / - Image buildup takes place in the background (back buffer) / - Visible image is independent of this (front buffer) / Rendering / Process
Commentary	This slide is about Double Buffering. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Use of two color buffers to avoid flicker effects / - Image buildup takes place in the background (back buffer) / - Visible image is independent of this (front buffer) / Rendering / Process
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Double Buffering'?
Page 86	2.7 Pixel-based Rendering
Source cue	Direct pixel processing without involving 3D models
Commentary	This slide is about 2.7 Pixel-based Rendering. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Direct pixel processing without involving 3D models

Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '2.7 Pixel-based Rendering' into a causal sentence instead of repeating the slide title?
Page 87	Pixel-based Rendering Operations
Source cue	- Generally / - Read and write operations for various image data formats, / usually as an external library (e.g., DevIl) / - Conversions between image data types / - In OpenGL done with framebuffer objects
Commentary	This slide is about Pixel-based Rendering Operations. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Generally / - Read and write operations for various image data formats, / usually as an external library (e.g., DevIl) / - Conversions between image data types / - In OpenGL done with framebuffer objects
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Pixel-based Rendering Operations'?
Page 88	Image Operations
Source cue	- Rendering image data with OpenGL / Copy Pixel Data / Draw/Read / Pixel / Geometry
Commentary	This slide is about Image Operations. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Rendering image data with OpenGL / Copy Pixel Data / Draw/Read / Pixel / Geometry
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Image Operations'?
Page 89	Write Pixels into Framebuffer
Source cue	- glDrawPixels(x, y, width, height, format, type, pixels); / - Example // 64x64 array with rgb components / GLubyte img[64][64][3]; / void draw() {
Commentary	This slide is about Write Pixels into Framebuffer. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - glDrawPixels(x, y, width, height, format, type, pixels); / - Example // 64x64 array with rgb components / GLubyte img[64][64][3]; / void draw() {
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Write Pixels into Framebuffer'?
Page 90	Read Pixels from Framebuffer

Source cue	- glReadPixels(x, y, width, height, format, type, pixels); / - Example / GLubyte* img = 0; / int width, height; // canvas size / void snapshot() {
Commentary	This slide is about Read Pixels from Framebuffer. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - glReadPixels(x, y, width, height, format, type, pixels); / - Example / GLubyte* img = 0; / int width, height; // canvas size / void snapshot() {
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Read Pixels from Framebuffer'?

Page 91	Formats & Data Types
Source cue	- Format information for pixel array operations / - GL_RGB – red-green-blue values of the frame buffer (FB) / - GL_RED – red values of the FB / - GL_ALPHA – transparency values of the FB / - GL_RGBA – red-Green-Blue-Alpha values of the FB
Commentary	This slide is about Formats & Data Types. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Format information for pixel array operations / - GL_RGB – red-green-blue values of the frame buffer (FB) / - GL_RED – red values of the FB / - GL_ALPHA – transparency values of the FB / - GL_RGBA – red-Green-Blue-Alpha values of the FB
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Formats & Data Types'?

Page 92	coordinate systems in the geometry stage
Source cue	- Rendering pipeline consist of geometry stage and rasterization stage / - Geometric primitives (mostly triangles) are transformed through different / coordinate systems in the geometry stage / - Rasterization results in fragments, which are pixel predecessors / - Fragments have to endure a sequence of tests in the rasterization stage, in
Commentary	This slide is about coordinate systems in the geometry stage. Read it as a data-flow explanation: scene or model data is processed step by step until valid framebuffer updates remain. The visible cue is: - Rendering pipeline consist of geometry stage and rasterization stage / - Geometric primitives (mostly triangles) are transformed through different / coordinate systems in the geometry stage / - Rasterization results in fragments, which are pixel predecessors / - Fragments have to endure a sequence of tests in the rasterization stage, in
Why it matters	Pipeline understanding lets you localize rendering errors instead of guessing randomly.
Check yourself	Can you name the input and output representation for 'coordinate systems in the geometry stage' in the rendering pipeline?

Page 93	Literature and other sources used in this chapter
---------	---

Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Literature and other sources used in this chapter. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?

Page 94	Guide: The Official Guide to Learning OpenGL
Source cue	- Text Books / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013. / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer
Commentary	This slide is about Guide: The Official Guide to Learning OpenGL. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Text Books / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013. / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Guide: The Official Guide to Learning OpenGL'?

02.1 Rendering Process

Source location: Section 2.1

Commentary

Object-based rendering starts with scene objects represented as primitives. The pipeline transforms vertices, assembles primitives, rasterizes them into fragments, and lets only some fragments become pixels. The important detail is that the representation changes at every stage.

The application stage prepares data and state. The geometry stage handles positions and primitives. The rasterization stage creates fragments and applies fragment operations. This separation helps you explain both theory questions and black-screen OpenGL bugs.

Mental model

Vertices become primitives, primitives become fragments, and fragments compete to become pixel updates.

Check yourself

What is the difference between a fragment and a final pixel?

02.2 OpenGL Overview

Source location: Section 2.2

Commentary

OpenGL is an API for controlling a graphics pipeline. It is not a renderer by itself in the sense of a single command that understands your scene. You create a context, provide buffers, set state, compile shaders, bind objects, and issue draw calls.

Modern OpenGL is stateful and shader-based. Many errors happen because the programmer assumes that a later call carries more information than it really does. The GPU uses the currently bound objects and current state at draw time.

Mental model

OpenGL draw calls consume the state machine as it exists right now.

Check yourself

Why can binding the wrong vertex array or shader program produce a valid draw call with wrong output?

02.3 OpenGL Rendering

Source location: Section 2.3

Commentary

Older OpenGL exposed a fixed-function style where many transformations and lighting choices were configured through predefined calls. Modern OpenGL expects you to write shader programs and explicitly manage data flow.

This is not just historical trivia. It explains why shader code, vertex attributes, uniforms, and buffer objects are central in the exercises. If the shader expects an attribute and the vertex layout does not provide it, the pipeline has no magic fallback.

Mental model

Modern OpenGL makes the data path explicit; that gives control but also creates responsibility.

Check yourself

Which data usually goes into vertex attributes, and which data usually goes into uniforms?

02.4 OpenGL Rendering Pipeline

Source location: Section 2.4

Commentary

The vertex shader runs per vertex and usually outputs clip-space position plus attributes for later interpolation. Primitive assembly groups vertices into points, lines, or triangles. Clipping handles geometry outside the view volume. Rasterization creates fragments.

The fragment shader computes candidate fragment output, often using interpolated attributes, uniforms, and textures. After that, tests and operations such as depth testing, stencil testing, blending, and masking decide

what reaches the framebuffer.

Mental model

Programmable shaders compute values; fixed pipeline stages decide coverage, interpolation, tests, and output rules.

Check yourself

If your geometry appears in wireframe but colors are wrong, which stages become suspicious?

02.5 Fragment Tests, Framebuffer, and Pixel-Based Rendering

Source location: Sections 2.5-2.7

Commentary

Fragment tests are the gatekeepers after shading. A fragment can have a perfectly valid color and still fail because depth, stencil, scissor, masking, or blending state prevents a visible update. This is why final pixels are a subset of generated fragments.

The framebuffer stores the final render targets: color buffers, depth buffers, stencil buffers, or custom attachments. Pixel-based rendering starts from image data instead of geometric primitives, which is useful for image processing but conceptually different from the object-to-pixel pipeline.

Mental model

The framebuffer is the destination; fragment operations are the rules for writing into it.

Check yourself

Why can two fragments with different colors produce only one visible pixel color?

End-of-lecture summary

If you remember only one thing from Lecture 02, remember this: This lecture turns the vague idea of rendering into a concrete sequence. It is the most important debugging map in the course: if an image is wrong, the pipeline tells you where the error could have entered.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.